

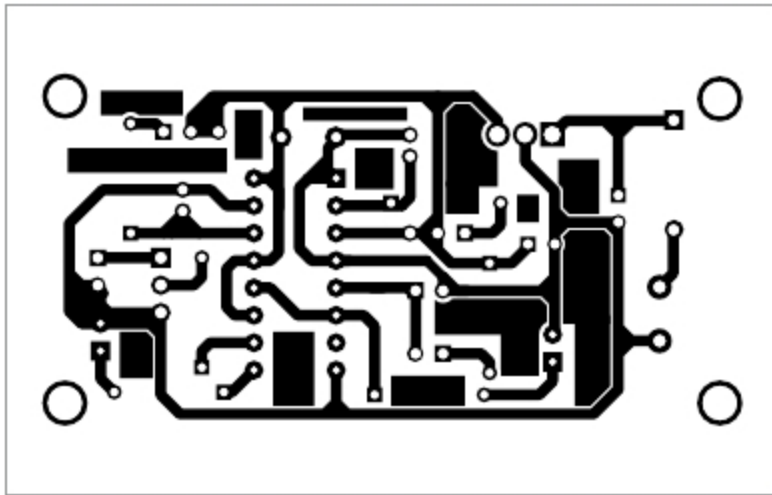


Fig. 3: Actual-size PCB layout for conveyor jam detector

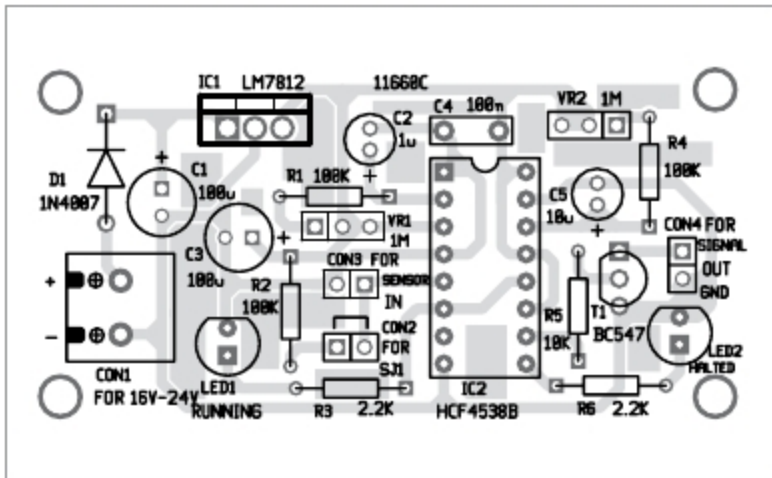


Fig. 4: Components layout of the PCB



Fig. 5: Idler wheel model

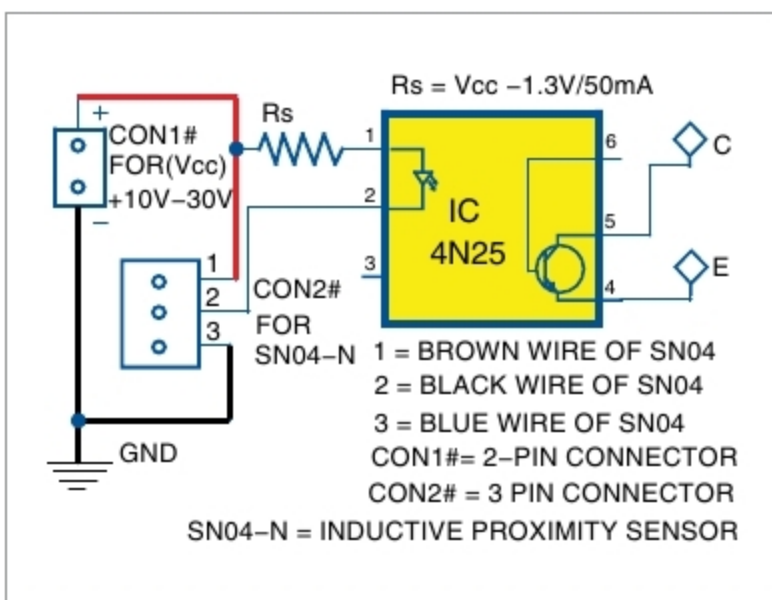


Fig. 6: Isolated interface circuit



Fig. 7: Partial view of author's prototype

100-kilo-ohm ($R1 + VR1$) to ensure that the first monostable is compatible with the sensor pulse input. The preferred pulse width of the second monostable is about ten times of the first monostable, which results in RC values as 100-kilo-ohm and $10\mu F$ (that is, $R4 + VR2 = 100\text{-kilo-ohm}$ and $C5 = 10\mu F$).

components $R1$, $VR1$, and $C1$ set the RC time constant of IC2A whereas $R4$, $VR2$, and $C5$ set RC time constant for IC2B.

The input pull-up resistor $R2$ is added intentionally because the circuit input at CON3 is configured to handle negative-going (falling edge) pulses only. For a sensor with an inbuilt pull-up resistor, simply remove shorting jumper (SJ1). Output pin 10 of the second monostable (IC2B) drives the output (CON4) via the NPN open-collector transistor switch (T1).

The DC input (16V to 24V) connected across CON1 is protected against reverse polarity feed by one 1N4007 diode (D1). The fixed voltage regulator LM7812 (IC1) lets you power up the circuit through any unregulated DC supply adaptor.

Calibration

Calibrate the setup with the help of potmeters $VR1$ and $VR2$. Assuming the rotation per minute (RPM) of the idler wheel is 600, the concerned rotation sensor assembly (RPM sensor) giving one pulse per rotation results in a period of 100ms.

Since $C1$ is a $1\mu F$ capacitor, you need a resistance value of

The calibration will ensure that while the conveyor is moving, the first monostable of IC2A is triggered by the sensor pulses through CON3, which in turn triggers the second monostable of IC2B. Hence, T1 gives a closed output through CON4. This action continues till the conveyor moves at the desired speed.

In case the conveyor comes to a halt (or slows down), IC2A no longer gets the trigger pulse, therefore T1 is opened by IC1B (after a short delay) to indicate the conveyor jam or failure.

Construction and testing

The actual-size PCB layout for the conveyor jam detector is shown in Fig. 3 and its components layout in Fig. 4.

The prototype was tested using a square type inductive proximity sensor SN04-N, powered by a 12V supply. The author's model of DC motor driven idler wheel used for testing the prototype is shown in Fig. 5.

Note that SN04-N has an internal 10-kilo-ohm pull-up resistor, so if you are powering it (or any other inductive/optical proximity/rotation sensor module with internal pull-up resistor) from an external power supply (usually 10V to 30V), remember to include a simple isolated interface circuit like the one shown in Fig. 6.

Partial view of author's prototype wired on a bread board is shown in Fig. 7. **EFY**



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